

Fundamentals of Elementary Particle Physics

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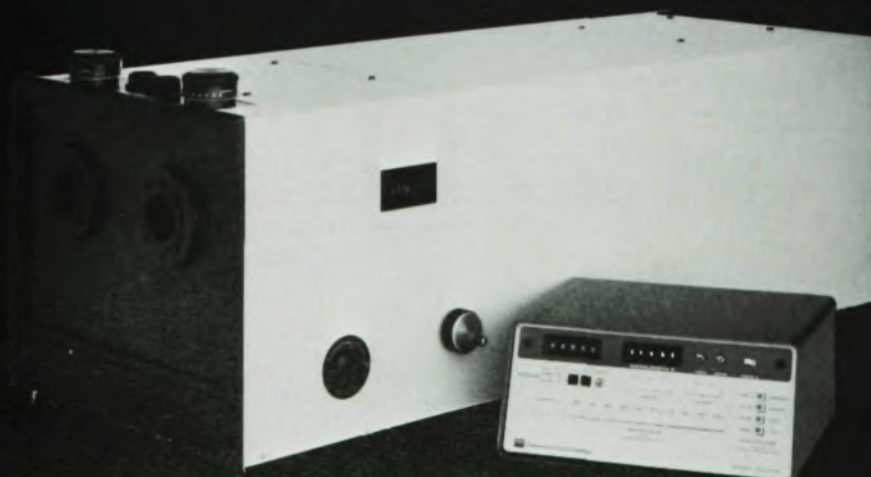
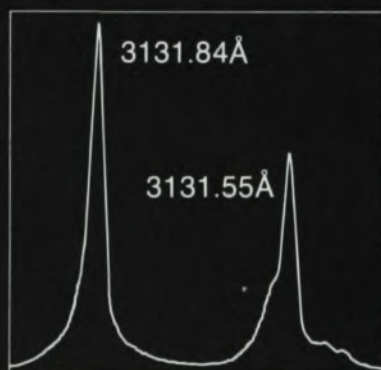
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*A. B. Shafer, L. R. Megill, and L. Droppleman, *J. Opt. Soc. Am.*, 54, 879-87 (1964)
J. Reader, *J. Opt. Soc. Am.*, 59, 1189-96 (1969)



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particularly lucid and satisfying manner. The formalism throughout the book is lightly sketched and carefully kept within the means of the undergraduate student. For example the Dirac equation is mentioned, but not written down; addition of angular momenta is left to an appendix; and the mathematics of group theory is briefly discussed but not employed. The treatment is correct, nonetheless, and the reader will come away with a useful understanding of the material. Tassie clearly explains concepts of virtual states, amplitudes and probabilities, renormalization and the statistics of bosons and fermions with an incisive blend of prose and mathematical formulae, while retaining throughout a proper treatment of particles and quantum-mechanical states.

There are some instances in which the treatment is not successful. A long chapter on resonances suffers badly from a confusing and inadequate discussion of kinematics. Phase space is referred to as "statistical theory" in lieu of a sound mathematical development, and an appendix on relativistic kinematics is devoted mainly to the Jacobian-peak missing-mass spectrometer. The section on Regge poles (which are discussed only as a means of classifying particles) is little more than a sampling of the jargon.

This book should prove a very serviceable text in that nearly all of the important topics are at least touched on, and most are treated with great clarity both of presentation and organization. The challenge is not fully met, however—the book is almost solely devoted to the classification and properties of the particles as we know them, with virtually no discussion of the central questions currently facing researchers in the field. Indeed, Tassie chooses not to deal at all with the collisions of strongly interacting particles, except in the context of resonance formation. Thus, although the intersecting storage rings at CERN are described, and an appendix lists the highest energies attained by various types of particle accelerators, the student is left with no clear notion of why such machines are built.

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Fundamentals of Elementary Particle Physics

M. J. Longo
226 pp., McGraw-Hill, New York
1973. \$9.95

Elementary-particle physics now encompasses such a large range of special

topics that it is quite a challenge to provide a brief, lucid, but not too superficial account of the field. Michael Longo has turned his undergraduate course notes into a rather well conceived and exciting survey of particle physics.

Longo is a high-energy experimentalist, and the viewpoint presented in the book is experimental and phenomenological in emphasis. The book ranges over topics in strong, electromagnetic and weak interactions, with special emphasis on a straightforward description of the relevant phenomena.

An informative chapter on detection devices and on high-energy experiments is followed by descriptions of famous experiments and discussions of their physical relevance. Although the accounts are brief, the "flavor" and style of high-energy physics is rather well communicated. Longo's enthusiasm for the subject clearly shows through, and the full-page pictures of famous physicists and experimental setups add to the attractiveness of the book.

The reader is expected to have some working knowledge of quantum mechanics. The oftentimes unconventional notation could be confusing for some students. A second difficulty with the book is the first chapter, where the quantum numbers, conservation laws, and particle spectrum are all unceremoniously dumped in the reader's lap. In spite of these technical shortcomings, this could be a useful elementary text for an undergraduate course on particle physics.

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Electron Optics and Electron Microscopy

P. W. Hawkes
244 pp. Barnes and Noble,
New York, 1972. \$17.00

The subject of electron optics, as it applies to the electron microscope, is a substantial one, with a very extensive literature to which this little volume is the latest addition.

The book contains chapters on the history of and justification for the electron microscope, electron lenses (including their most common aberrations), the electron microscope itself, the scanning microscope and microanalyzer and a short chapter on applications. This global view of the electron microscope is compressed into a little over 200 pages. It can be appreciated, then, that most of the material is in a very condensed form, and many aspects are either glossed over or completely omitted. Nevertheless, the

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